

## Xiaoling Hu

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**Current Position** • **Harvard Medical School, Athinoula A. Martinos Center for Biomedical Imaging, USA** Aug. 2023 - Present  
*Postdoctoral Research Fellow*  
- Hosted by Prof. Juan Eugenio Iglesias and Prof. Bruce Fischl

**Research Interests** My research interests are in machine learning, computer vision, and their applications in healthcare, multimodal AI, generative AI, and trustworthy learning systems. In particular, I am interested in:

- **Topological Deep Learning**
- **Trustworthy Machine Learning**
- **Multimodal AI and Generative AI (GenAI)**
- **VLM/LLM Reasoning and Efficiency**
- **Healthcare Applications**

**Education** • **Stony Brook University, Department of CS, USA** Jan. 2018 - June 2023  
*Doctor of Philosophy*  
- Advisor: Chao Chen  
- Thesis: Learning Topological Representations for Deep Image Understanding  
- Committee: Chao Chen, Dimitris Samaras, Haibin Ling, Li Fuxin

• **Tsinghua University, Department of EE, China** Sep. 2014 - June 2017  
*Master of Engineering*

• **Huazhong University of Science and Technology, Department of EE, China** Sep. 2010 - June 2014  
*Bachelor of Science*

**Selected Publications** (\* indicates equal contribution, \_ denotes students (co-)mentored by me, ‡ denotes (co)-senior supervision)

- [1] **Act Like a Pathologist: Tissue-Aware Whole Slide Image Reasoning**  
Wentao Huang, Weimin Lyu, Peiliang Lou, Qingqiao Hu, **Xiaoling Hu**, Shahira Abousamra, Wenchao Han, Ruifeng Guo, Jiawei Zhou, Chao Chen, Chen Wang  
*The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026
- [2] **RankByGene: Gene-Guided Histopathology Representation Learning Through Cross-Modal Ranking Consistency**  
Wentao Huang, Meilong Xu, **Xiaoling Hu**, Shahira Abousamra, Aniruddha Ganguly, Saarthak Kapse, Alisa Yurovsky, Prateek Prasanna, Tahsin Kurc, Joel Saltz, Michael L. Miller, Chao Chen  
Major revision for *IEEE Transactions on Medical Imaging (TMI)*
- [3] **MATCH: Multi-faceted Adaptive Topo-Consistency for Semi-Supervised Histopathology Segmentation**  
Meilong Xu\*, **Xiaoling Hu**\*, Shahira Abousamra, Chen Li, Chao Chen

*Thirty-Ninth Conference on Neural Information Processing Systems (NeurIPS)*, 2025  
Short version is selected as **Oral Presentation** at Imageomics Workshop at NeurIPS 2025

- [4] **Text-Driven Weakly Supervised OCT Lesion Segmentation with Structural Guidance**  
Jiaqi Yang, Nitish Mehta, **Xiaoling Hu**, Chao Chen, Chia-Ling Tsai  
*The Journal of Biomedical and Health Informatics (JBHI)*, 2025
- [5] **Learn2Synth: Learning Optimal Data Synthesis Using Hypergradients for Brain Image Segmentation**  
**Xiaoling Hu**, Xiangrui Zeng, Oula Puonti, Juan Eugenio Iglesias, Bruce Fischl<sup>‡</sup>, Yaël Balbastre<sup>‡</sup>  
*International Conference on Computer Vision (ICCV)*, 2025
- [6] **TopoCellGen: Generating Histopathology Cell Topology with a Diffusion Model**  
Meilong Xu, Saumya Gupta, **Xiaoling Hu**, Chen Li, Shahira Abousamra, Dimitris Samaras, Prateek Prasanna, Chao Chen  
*The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025 (**Oral Presentation, acceptance rate < 1%**)  
Short version is also presented at CVPR 2025 Workshop GMCV
- [7] **Hierarchical Uncertainty Estimation for Learning-based Registration in Neuroimaging**  
**Xiaoling Hu**, Karthik Gopinath, Peirong Liu, Malte Hoffmann, Koen Van Leemput, Oula Puonti<sup>‡</sup>, Juan Eugenio Iglesias<sup>‡</sup>  
*International Conference on Learning Representations (ICLR)*, 2025
- [8] **Semi-supervised Segmentation of Histopathology Images with Noise-Aware Topological Consistency**  
Meilong Xu, **Xiaoling Hu**, Saumya Gupta, Shahira Abousamra, Chao Chen  
*European Conference on Computer Vision (ECCV)*, 2024
- [9] **Brain-ID: Learning Contrast-agnostic Anatomical Representations for Brain Imaging**  
Peirong Liu, Oula Puonti, **Xiaoling Hu**, Daniel C. Alexander, Juan Eugenio Iglesias  
*European Conference on Computer Vision (ECCV)*, 2024
- [10] **Registration by Regression (RbR): a framework for interpretable and flexible atlas registration**  
Karthik Gopinath\*, **Xiaoling Hu\***, Malte Hoffmann, Oula Puonti<sup>‡</sup>, Juan Eugenio Iglesias<sup>‡</sup>  
*Workshop on Biomedical Image Registration-MICCAI (WBIR)*, 2024
- [11] **P-Count: Persistence-based Counting of White Matter Hyperintensities in Brain MRI**  
**Xiaoling Hu**, Annabel Sorby-Adams, Frederik Barkhof, William Kimberly, Oula Puonti, Juan Eugenio Iglesias  
*Workshop on Topology- and Graph-Informed Imaging Informatics-MICCAI (TGI3)*, 2024
- [12] **Semi-Supervised Contrastive VAE for Disentanglement of Digital Pathology Images**  
Mahmudul Hasan, **Xiaoling Hu**, Shahira Abousamra, Prateek Prasanna, Joel Saltz, Chao Chen

*International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024

- [13] **Hard Negative Sample Mining for Whole Slide Image Classification**  
Wentao Huang, **Xiaoling Hu**, Shahira Abousamra, Prateek Prasanna, Chao Chen  
*International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024
- [14] **Spatial Diffusion for Cell Layout Generation**  
Chen Li, **Xiaoling Hu**, Shahira Abousamra, Meilong Xu, Chao Chen  
*International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024
- [15] **Anomaly-guided weakly supervised lesion segmentation on retinal OCT images**  
Jiaqi Yang, Nitish Mehta, Gozde Merve Demirci, **Xiaoling Hu**, Meera Ramakrishnan, Mina Naguib, Chao Chen, Chialing Tsai  
*Medical Image Analysis (MedIA)*, 2024
- [16] **Topology-Aware Uncertainty for Image Segmentation**  
Saumya Gupta, Yikai Zhang, **Xiaoling Hu**, Prateek Prasanna, Chao Chen  
*Thirty-Seventh Conference on Neural Information Processing Systems (NeurIPS)*, 2023
- [17] **Calibrating Uncertainty for Semi-Supervised Crowd Counting**  
Chen Li, **Xiaoling Hu**, Shahira Abousamra, Chao Chen  
*International Conference on Computer Vision (ICCV)*, 2023
- [18] **Enhancing Modality-Agnostic Representations via Meta-Learning for Brain Tumor Segmentation**  
Aishik Konwer, **Xiaoling Hu**, Xuan Xu, Joseph Bae, Chao Chen, Prateek Prasanna  
*International Conference on Computer Vision (ICCV)*, 2023
- [19] **Learning Probabilistic Topological Representations Using Discrete Morse Theory**  
**Xiaoling Hu**, Dimitris Samaras, Chao Chen  
*International Conference on Learning Representations (ICLR)*, 2023 (**Spotlight Presentation, notable-top-25% of accepted papers**)  
*Short version is selected as Oral Presentation at Medical Imaging meets NeurIPS Workshop, 2023*
- [20] **Confidence Estimation Using Unlabeled Data**  
Chen Li, **Xiaoling Hu**, Chao Chen  
*International Conference on Learning Representations (ICLR)*, 2023
- [21] **Structure-Aware Image Segmentation with Homotopy Warping**  
**Xiaoling Hu**  
*Thirty-Sixth Conference on Neural Information Processing Systems (NeurIPS)*, 2022
- [22] **Learning Topological Interactions for Multi-Class Medical Image Segmentation**  
Saumya Gupta\*, **Xiaoling Hu\***, James Kaan, Michael Jin, Mutshipay Mpoy, Katherine Chung, Gagandeep Singh, Mary Saltz, Tahsin Kurc, Joel Saltz, Apostolos Tsiaopoulos, Prateek Prasanna, Chao Chen  
*European Conference on Computer Vision (ECCV)*, 2022 (**Oral Presentation, acceptance rate 2.7%**)

- [23] **Trigger Hunting with a Topological Prior for Trojan Detection**  
**Xiaoling Hu**, Xiao Lin, Michael Cogswell, Yi Yao, Susmit Jha, Chao Chen  
*International Conference on Learning Representations (ICLR)*, 2022
- [24] **A Manifold View of Adversarial Risk**  
Wenjia Zhang, Yikai Zhang, **Xiaoling Hu**, Mayank Goswami, Chao Chen, Dimitris Metaxas  
*International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2022
- [25] **Topology-Attention ConvLSTM Network for 3D Image Segmentation**  
Jiaqi Yang\*, **Xiaoling Hu\***, Chao Chen, Chialing Tsai  
*International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2021
- [26] **Topology-Aware Segmentation Using Discrete Morse Theory**  
**Xiaoling Hu**, Yusu Wang, Li Fuxin, Dimitris Samaras, Chao Chen  
*International Conference on Learning Representations (ICLR)*, 2021 (**Spotlight Presentation, acceptance rate 5.6%**)
- [27] **3D Topology-Preserving Segmentation with Compound Multi-Slice Representation**  
Jiaqi Yang\*, **Xiaoling Hu\***, Chao Chen, Chialing Tsai  
*IEEE International Symposium on Biomedical Imaging (ISBI)*, 2021
- [28] **Topology-Preserving Deep Image Segmentation**  
**Xiaoling Hu**, Li Fuxin, Dimitris Samaras, Chao Chen  
*Thirty-Third Conference on Neural Information Processing Systems (NeurIPS)*, 2019

## Preprints

- [1] **Learning to Upscale 3D Segmentations in Neuroimaging**  
**Xiaoling Hu**, Peirong Liu, Dina Zemlyanker, Jonathan Williams Ramirez, Oula Puonti, Juan Eugenio Iglesias  
*Tech Report*, 2025
- [2] **Topo-R1: Detecting Topological Anomalies via Vision-Language Models**  
Meilong Xu, Qingqiao Hu, **Xiaoling Hu**, Shahira Abousamra, Xin Yu, Weimin Lyu, Kehan Qi, Dimitris Samaras, Chao Chen  
*Tech Report*, 2026
- [3] **Unrolled Networks are Conditional Probability Flows in MRI Reconstruction**  
Kehan Qi, Saumya Gupta, **Xiaoling Hu**, Qingqiao Hu, Weimin Lyu, Chao Chen  
*Tech Report*, 2025
- [4] **RB-FT: Rationale-Bootstrapped Fine-Tuning for Video Classification**  
Meilong Xu, Di Fu, Jiaying Zhang, Gong Yu, Jiayu Zheng, **Xiaoling Hu**, Dongdi Zhao, Feiyang Li, Chao Chen, Yong Cao  
*Tech Report*, 2025
- [5] **LoC-Path: Learning to Compress for Pathology Multimodal Large Language Models**  
Qingqiao Hu, Weimin Lyu, Meilong Xu, Kehan Qi, **Xiaoling Hu**, Saumya Gupta, Jiawei Zhou, Chao Chen  
*Tech Report*, 2025

- [6] **Uncertainty Estimation for Pretrained Medical Image Registration Models via Transformation Equivariance**  
Lin Tian, **Xiaoling Hu**, Juan Eugenio Iglesias  
*Tech Report*, 2025
- [7] **Bézier Meets Diffusion: Robust Generation Across Domains for Medical Image Segmentation**  
Chen Li, Meilong Xu, **Xiaoling Hu**, Weimin Lyu, Chao Chen  
*Tech Report*, 2025
- [8] **Adversarial Vessel-Unveiling Semi-Supervised Segmentation for Retinopathy of Prematurity Diagnosis**  
Gozde Merve Demirci, Jiachen Yao, Ming-Chih Ho, **Xiaoling Hu**, Wei-Chi Wu, Chao Chen, and Chia-Ling Tsai  
*Tech Report*, 2024
- [9] **Deep Statistic Shape Model for Myocardium Segmentation**  
**Xiaoling Hu**, Xiao Chen, Terrence Chen, Shanhui Sun  
*Tech Report*, 2022

**Selected  
Honors and  
Awards**

- Catacosinos Fellowship (2 out of 200+ PhD students in SBU CS Department), 2023
- NeurIPS Travel Award, 2019
- First-class Scholarship, Tsinghua University, 2016 (5%)

**Mentoring**

- Chen Li (**ICLR'23, ICCV'23, MICCAI'24**), Ph.D. student at Department of BMI, Stony Brook University  
Since Fall 2021
- Meilong Xu (**ECCV'24, CVPR'25 Oral Presentation, NeurIPS'25**), Ph.D. student at Department of CS, Stony Brook University  
Since Summer 2023
- Wentao Huang (**MICCAI'24, CVPR'26, TMI major revision**), Ph.D. student at Department of CS, Stony Brook University  
Since Summer 2023
- Qingqiao Hu, Ph.D. student at Department of CS, Stony Brook University  
Since Fall 2024
- Kehan Qi, Ph.D. student at Department of BMI, Stony Brook University  
Since Fall 2025
- Jiaqi Yang (**MICCAI'21, ISBI'21, MedIA'24, JBHI'25**), Ph.D. student at Department of CS, CUNY  
Spring 2020 – Summer 2023
- Saumya Gupta (**ECCV'22 Oral Presentation, NeurIPS'23**), Ph.D. student at Department of CS, Stony Brook University  
Fall 2021 – Summer 2023
- Mahmudul Hasan (**MICCAI'24**), Ph.D. student at Department of CS, Stony Brook University  
Summer 2023 – Summer 2024
- Luca Drole, Master student in BME, ETH Zurich  
Spring - Fall 2025
- John Xie, High School student → University of Michigan  
Summer 2021

**Professional  
Service**

**Organizer**

- MICCAI'25 workshop on *Topology- and Graph-Informed Imaging Informatics (TGI3)* 2025
- MICCAI'24 workshop on *The First Workshop on Topology- and Graph-Informed Imaging Informatics (TGI3)* 2024
- MICCAI'23 tutorial on *Topology-Driven Image Analysis* 2023

**Area Chair**

- International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) 2025, 2026
- Conference on Neural Information Processing Systems (NeurIPS) 2025, 2026
- The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026
- International Conference on Artificial Intelligence and Statistics (AISTATS) 2026

**Reviewing**

- International Conference on Machine Learning (ICML) Since 2022
- International Conference on Learning Representations (ICLR) Since 2022
- Conference on Neural Information Processing Systems (NeurIPS) 2021 - 2024
- Computer Vision and Pattern Recognition (CVPR) 2021 - 2025
- European Conference on Computer Vision (ICCV) Since 2021
- European Conference on Computer Vision (ECCV) Since 2022
- Winter Conference on Applications of Computer Vision (WACV) Since 2022
- International Conference on Artificial Intelligence and Statistics (AISTATS) 2022 - 2025
- Learning on Graphs Conference (LoG) 2022 - 2025
- Medical Imaging with Deep Learning (MIDL) Since 2022
- AAAI Conference on Artificial Intelligence (AAAI) Since 2022
- International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) 2020 - 2024
- Transactions on Machine Learning Research (TMLR)
- IEEE Transactions on Medical Imaging (TMI)
- Medical Image Analysis (MedIA)

**Talks**

**Principled Learning for Medical AI: Structure, Reliability, and Interpretability**

- HIT Webinar - *Healthcare, Intelligence, Technology* Aug. 2025
- Department of CSE, University of North Texas Nov. 2025

***Learn2Synth*: A Learnable Data Synthesis Strategy for Image Segmentation**

- Nobrainer Seminar, Massachusetts Institute of Technology June 2024

## Deep Structural Reasoning for Biomedical Imaging

- School of CAI, Arizona State University Feb. 2024

## Topology-Aware Deep Image Segmentation

- MICCAI'23 tutorial on *Topology-Driven Image Analysis*, Vancouver Oct. 2023

## Learning Topological Representations for Deep Image Understanding

- Department of CS, Florida State University Apr. 2023
- Department of BMI, Ohio State University Mar. 2023
- Department of CS, Rochester Institute of Technology Feb. 2023
- Department of ECE, University of California, Riverside Feb. 2023
- Athinoula A. Martinos Center for Biomedical Imaging, Massachusetts General Hospital & Harvard Medical School Nov. 2022

## Learning Probabilistic Topological Representations Using Discrete Morse Theory

- Medical Imaging meets NeurIPS Workshop, New Orleans Dec. 2022

## Topology-Informed Image Analysis

- Center for Computational Neuroscience, Flatiron Institute Oct. 2022

## Topology-Aware Deep Image Segmentation

- Geometry and Topology meet Data Analysis and Machine Learning Aug. 2021

## Topology-aware Segmentation Using Discrete Morse Theory

- International Conference on Learning Representations (ICLR) May 2021

## Industry Experiences

- **Allen Institute, USA** May 2022 - Aug. 2022  
*Research Intern*  
Mentor: *Dr. Matheus Viana*  
Topic: Topology-Aware Image Segmentation
- **United Imaging Intelligence (UII), USA** May 2021 - Aug. 2021  
*Research Intern*  
Mentor: *Dr. Shanhui Sun*  
Topic: Deep Shape Model Based Network
- **Tencent Youtu Lab, China** Jun. 2017 - Jan. 2018  
*Research Intern*  
Mentor: *Dr. Yuwing Tai*  
Topic: Clothes Detection, Attribute Prediction

## References

- **Chao Chen**  
Associate Professor, Stony Brook University  
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<https://chaochen.github.io/>

- **Juan Eugenio Iglesias**  
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<https://lemon.martinos.org/pi/>
- **Bruce Fischl**  
Professor, Massachusetts General Hospital & Harvard Medical School  
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<https://scholar.google.com/citations?user=t7mytXkAAAAJ&hl=en>
- **Dimitris Samaras**  
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- **Fuxin Li**  
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<https://web.engr.oregonstate.edu/~lif/>
- **Prateek Prasanna**  
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<https://you.stonybrook.edu/imaginelab/>